BỘ GIÁO DỤC VÀ ĐÀO TẠO

TRƯỜNG ĐẠI HỌC MỞ THÀNH PHỐ HỒ CHÍ MINH

Nâng cao tính tự nhiên và tính ngữ cảnh trong tương tác hội thoại với NPC bằng mô hình ngôn ngữ tinh chỉnh được thông tin bởi trạng thái thế giới game động trong game tường thuật

Mã số đề tài: 866

Chủ nhiệm đề tài: Lê Trần Minh Phúc

Khoa: Đào Tạo Đặc Biệt

Người hướng dẫn: TS. Lê Đăng Giáp

TP. Hồ Chí Minh, 2026

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TRƯỜNG ĐẠI HỌC MỞ THÀNH PHỐ HỒ CHÍ MINH	**Độc lập - Tự do - Hạnh phúc**

THÔNG TIN KẾT QUẢ NGHIÊN CỨU CỦA ĐỀ TÀI

1. Thông tin chung:

- Tên đề tài: Nâng cao tính tự nhiên và tính ngữ cảnh trong tương tác hội thoại với NPC bằng mô hình ngôn ngữ tinh chỉnh được thông tin bởi trạng thái thế giới game động trong game tường thuật
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2. Mục tiêu đề tài:

Xây dựng và đánh giá một hệ thống hội thoại NPC tích hợp đầu-cuối cho trò chơi tường thuật, trong đó mô hình ngôn ngữ cục bộ được điều kiện hóa theo trạng thái trò chơi động nhằm nâng cao độ bám ngữ cảnh, tính nhất quán vai trò nhân vật và chất lượng phản hồi trong điều kiện triển khai thời gian thực.

3. Tính mới và sáng tạo:

Đề tài kết hợp đồng thời trích xuất ngữ cảnh từ UE5, truy hồi lai có cơ chế guard chống poisoning/trust-spoof, sinh phản hồi bằng mô hình tinh chỉnh cục bộ và lớp kiểm soát hậu sinh có cơ chế fallback trong cả hai nhánh triển khai Python và C++. Thiết kế này cho phép đồng nhất hành vi giữa môi trường đánh giá và runtime trong game, đồng thời hỗ trợ phân tích ablation theo từng thành phần.

4. Kết quả nghiên cứu:

Trên bộ 112 kịch bản cố định, cấu hình kiểm soát theo ngữ cảnh cải thiện có ý nghĩa thống

kê so với cấu hình ngữ cảnh thô ở các chỉ số chính: context relevance (+0.2131), persona consistency (+0.2150), naturalness (+0.1158) và overall quality (+0.1808). Trong kiểm thử đối kháng, tỷ lệ tấn công thành công giảm từ 1.0 xuống 0.0 dưới cấu hình guard.

5. Đóng góp về mặt kinh tế - xã hội, giáo dục và đào tạo, an ninh, quốc phòng và khả năng áp dụng của đề tài:

Kết quả đề tài đóng góp quy trình kỹ thuật có thể kiểm chứng cho phát triển NPC hội thoại trong sản phẩm trò chơi số, hỗ trợ đào tạo liên ngành giữa AI và phát triển game, đồng thời nhấn mạnh cơ chế kiểm soát rủi ro nội dung khi ứng dụng mô hình ngôn ngữ trong hệ thống tương tác.

6. Công bố khoa học của sinh viên từ kết quả nghiên cứu của đề tài:

Hiện chưa có công bố khoa học chính thức; kết quả đang được hoàn thiện dưới dạng bản thảo học thuật và báo cáo kỹ thuật nội bộ.

TP. Hồ Chí Minh, 2026

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List of abbreviations

Abbreviation	Description
ASR	Attack Success Rate
BM25	Best Matching 25 retrieval model
CI	Confidence Interval
DPO	Direct Preference Optimization
LLM	Large Language Model
MRR	Mean Reciprocal Rank
nDCG	Normalized Discounted Cumulative Gain
NPC	Non-Player Character
OQ	Overall Quality
QLoRA	Quantized Low-Rank Adaptation
RAG	Retrieval-Augmented Generation
RQ	Research Question
TTFT	Time to First Token
UE5	Unreal Engine 5

Abstract

This paper presents an end-to-end system for narrative-game NPC dialogue that couples fine-tuned local language models with live game-state conditioning. The goal is to improve naturalness, persona consistency, and contextual relevance while preserving deployability in Unreal Engine 5. The architecture integrates four tightly coupled components: context extraction from game state, hybrid retrieval with poisoning-aware guard scoring, local response generation, and post-generation response control deployed in both Python and C++ runtime paths.

Evaluation is conducted with fixed scenarios, paired significance testing, confidence intervals, ablations, and a completed multi-rater study. On a 112-scenario benchmark, controlled contextual generation outperforms raw contextual generation with mean deltas of +0.2131 (context relevance), +0.2150 (persona consistency), +0.1158 (naturalness), and +0.1808 (overall quality), with one-sided bootstrap $p(\Delta \leq 0) = 0.0$ across core metrics. Against external no-context baselines (phi3:mini and phi3:latest), the controlled system achieves statistically positive outcomes on 10 of 12 tested metrics per baseline and improves overall quality in both comparisons.

Human evaluation (324 ratings) reports substantial preference for the controlled system and moderate inter-rater agreement ($\kappa = 0.5329$ mean pairwise). Security stress tests show complete suppression of successful poisoning-follow behavior under the guard configuration (attack success rate reduced from 1.0 to 0.0). The primary negative finding is serving efficiency: latency and throughput are not superior to lighter baseline serving configurations. The main contribution is a reproducible, evidence-scoped framework that demonstrates quality and robustness gains without obscuring performance trade-offs.

Keywords

NPC dialogue, game AI, local LLM, UE5, QLoRA, response control, retrieval robustness, benchmark reproducibility, neuro-symbolic systems.

Chapter 1: Problem context and research scope

1.1 Motivation and problem context

Narrative games require NPC dialogue that is simultaneously fluent, role-consistent, and grounded in dynamic world state. Survey evidence shows that classical authored dialogue pipelines remain dependable in production but incur rapidly growing authoring cost as interaction breadth increases [2, 3]. LLM-driven NPC systems improve linguistic flexibility, yet often fail under strict gameplay constraints such as quest-state consistency, inventory grounding, and stable persona control across multi-turn interaction [1, 2, 3].

The core challenge is therefore not text generation alone; it is runtime integration. A model can produce plausible standalone responses and still behave unreliably in an engine loop when context extraction, retrieval ranking, and post-generation control are weakly coupled. Prior work demonstrates promising qualitative behavior, but system-level evidence is often limited by narrow scenario coverage, weak adversarial testing, or insufficient statistical reporting [2, 4].

1.2 Research subject and system focus

This paper addresses that gap by designing and evaluating an end-to-end local runtime for context-grounded NPC dialogue in Unreal Engine 5. The method combines six components: bounded game-state serialization, prompt compilation, hybrid retrieval, guard-aware evidence scoring, local generation, and response control with fallback logic. The evaluation protocol is designed to answer a scientific question rather than present isolated demos: under fixed scenarios and controlled comparisons, does this architecture improve context relevance, persona consistency, and robustness without hiding negative trade-offs?

1.3 Scope and study setting

The study is scoped to narrative-game dialogue under UE5 runtime constraints, with local model serving and fixed-scenario evaluation under matched prompts. The experimental window includes both adversarial stress settings and a predefined human-rating

campaign so that automatic and human evidence are interpreted within the same protocol.

1.4 Methodological overview

To maintain scientific clarity, conclusions are explicitly scoped to measured axes. The study includes fixed-arm benchmarking, paired bootstrap tests, confidence intervals, ablations, human-rater agreement analysis, and adversarial stress evaluation. This allows us to report both improvements (quality and security robustness) and limitations (serving latency/throughput trade-offs).

1.5 Scholarly significance and contributions

Relative to prior LLM-in-games studies, the main contribution is not merely local fine-tuning, but the explicit coupling between architectural design and inferential evidence. The manuscript therefore explains why each component is included, how it is evaluated, and which effects remain after ablation. This structure helps separate empirically supported engineering claims from open hypotheses that require further study.

1.6 Research questions

The evaluation is organized around four explicit research questions. RQ1 examines whether controlled context-grounded inference improves context relevance, persona consistency, naturalness, and overall quality over raw contextual generation under matched prompts and decoding settings. RQ2 tests whether the same architecture outperforms no-context external baselines under identical scenario protocols. RQ3 asks whether guard-aware retrieval materially reduces adversarial follow-through under poisoned and trust-spoofed evidence conditions. RQ4 characterizes the quality-latency trade-offs that emerge when guard and response-control mechanisms are enabled in an in-engine runtime.

1.7 Objectives and formal problem statement

Formally, given player input x_t, dynamic world state s_t, retrieval corpus $\mathcal{D}$, and NPC persona profile p, the objective is to generate response y_t such that:

$$\max Q(y_t; x_t, s_t, p) \tag{1}$$

$$\text{s.t. } R(y_t, s_t) \leq \epsilon_r,\ C(y_t, p) \geq \tau_c,\ L(y_t) \leq \tau_l, \tag{2}$$

where Q aggregates quality metrics, R denotes safety/risk violations, C denotes persona/context constraints, and L denotes runtime constraints.

For repeated interactions, the policy objective is:

$$\pi^\star = \arg\max_{\pi}\ \mathbb{E}_{(x_t,s_t,p)\sim\mathcal{T}}\left[Q(\pi(x_t, s_t, p))\right], \tag{3}$$

with quality decomposed as:

$$\begin{aligned} Q &= w_{cr}CR + w_{pc}PC + w_n N + w_b B - w_v V, \\ \sum_i w_i &= 1, \quad w_i \geq 0, \end{aligned} \tag{4}$$

where CR is context relevance, PC is persona consistency, N is naturalness, B is semantic similarity score, and V is violation risk. The empirical program then addresses four linked questions: whether controlled context-grounded inference improves over raw contextual generation, whether it outperforms no-context external baselines under identical protocols, whether guard-aware retrieval reduces adversarial follow-through, and what serving trade-offs emerge when stronger control mechanisms are enabled. The contributions are therefore architectural, methodological, and empirical: an integrated UE5 runtime that combines state extraction, guarded hybrid retrieval, local generation, and runtime response control; parity between Python and C++ control paths; a fixed-arm benchmark with inferential statistics and multi-rater evaluation; and a reproducibility package that preserves both positive and negative findings.

For chapter-level traceability, Chapter 2 develops the theoretical and comparative grounding for these objectives, Chapter 3 defines the system and evaluation procedures

used to test them, Chapter 4 reports quantitative and qualitative findings under that protocol, and Chapter 5 provides bounded conclusions and deployment-oriented recommendations.

Chapter 2: Literature foundation and conceptual framing

2.1 International research landscape

Rule-based game dialogue systems remain strong in predictability and safety under strict authored quest logic, but their principal limitation is scalability because open-ended interaction breadth requires high authoring and maintenance cost [2, 3]. LLM-driven NPC systems invert that trade-off by improving fluency and coverage, but they introduce instability under dynamic state constraints and longer interaction horizons [1, 2, 3]. Recent interaction studies report promising collaboration and immersion signals while still documenting unresolved challenges in response latency, deterministic grounding, and robust runtime control [4, 5]. As a result, the central open problem has shifted from linguistic plausibility to controllable integration under game-runtime constraints.

Parallel work on game-capable language agents broadens this picture. Voyager demonstrates strong open-world progression in Minecraft through curriculum generation, skill-library reuse, and iterative code refinement [12]. However, its objective is general embodied task completion rather than role-faithful NPC dialogue; consequently, it does not directly address persona consistency, narrative constraints, or dialogue-specific human preference evaluation. Similarly, SIMA emphasizes language-conditioned action execution across multiple 3D environments and shows progress on cross-world instruction following [13], but it is not designed as a narrative dialogue controller and therefore leaves unresolved the question of turn-level conversational admissibility under dynamic story state. At the interaction-interface level, voice-driven NPC systems improve perceived freedom and engagement [14], yet current designs often map free-form user input back to bounded authored options, which limits full generative adaptability under adversarial or contradictory context.

2.2 Domestic research landscape

In the Vietnamese context, the most sustained research infrastructure has been the VLSP community, which emphasizes shared tasks, benchmark construction, and reproducible evaluation pipelines across language and speech problems [6, 7]. This ecosystem has recently expanded toward large-model development for Vietnamese, including foundation-model adaptation and language-specific continual pretraining such as VinaLLaMA and Vi-Mistral-X [8, 9]. In parallel, early Vietnamese RAG and multimodal-LLM efforts have improved resource availability through open datasets, evaluation suites, and model releases [10, 11].

However, these domestic advances are concentrated in general NLP capability, benchmark engineering, or modality expansion, rather than in in-engine NPC dialogue systems with dynamic game-state grounding, adversarial retrieval controls, and deployment-parity validation. In other words, Vietnam has strong momentum in language technology building blocks, but still limited peer-reviewed system studies that integrate those components into a single, auditable narrative-game runtime. This is the specific domestic gap targeted by the present study.

2.3 Core constructs and operational definitions

This work uses four central concepts: context relevance as fidelity to dynamic world state, persona consistency as adherence to NPC role profile, naturalness as linguistic plausibility under interaction constraints, and robustness as resistance to malicious or inconsistent evidence injection.

2.4 Theoretical and methodological foundations

At the model-adaptation level, prompt engineering and instruction tuning provide fast behavior shaping [15, 16], while RLHF-style methods and preference optimization improve adherence to desired responses [17, 38]. Prompting strategies that explicitly structure intermediate reasoning and action feedback, such as Chain-of-Thought and ReAct, further strengthen controllability in complex task settings [21, 22]. Parameter-efficient adap-

tation through LoRA and QLoRA enables local deployment under constrained memory budgets [18, 19]. Retrieval-aware generation frameworks such as Self-RAG further show that adaptive retrieval and self-critique can improve factuality [20]. Their main weakness for deployment-critical NPC systems is that they prioritize general factual quality and controllability, but do not explicitly target adversarial evidence handling in game-specific retrieval pipelines.

Benchmarking work on LLM agents and long-context reasoning clarifies why these gaps persist in runtime game deployments. AgentBench shows that even strong models degrade on multi-turn decision tasks requiring sustained planning and instruction fidelity [36]. LongBench further shows that performance drops substantially as context length and task heterogeneity increase, and that retrieval-based compression only partially closes the gap for weaker long-context models [37]. For dynamic NPC dialogue, these findings imply that naive context accumulation is insufficient; bounded state serialization, retrieval prioritization, and explicit post-generation control are required to maintain stable behavior as interaction horizons expand.

Retrieval-augmented generation strengthens factual grounding, from the original RAG formulation to more recent survey syntheses and graph-based extensions [42, 44, 43, 49, 48]. However, retrieval also creates a new control surface where poisoning, trust spoofing, and ranking manipulation can alter downstream generation. Recent high-impact attack studies reinforce this concern: indirect prompt injection can compromise application behavior through retrieved context [45], knowledge-corruption attacks can induce attacker-chosen outputs with a small number of injected passages [46], and adversarial instructional prompts can subvert retrieval behavior even when user queries are fixed [47]. Many evaluations still emphasize benign relevance quality and under-report adversarial behavior. This is a key weakness for game NPC settings, where retrieval errors can directly induce in-character policy violations. Evaluation methodology presents a related issue: overlap-only scoring is insufficient for interactive dialogue quality, dialog-metric behavior can vary substantially across datasets and settings [24], and modern embedding- and judge-based protocols can still carry positional or verbosity biases [23, 30, 29, 31]. For this reason, interactive dialogue systems require

convergent evidence across automatic metrics, inferential statistics, human preference, and agreement analysis [50].

2.5 Hypotheses and analytical model

The integrated theoretical position tested in this manuscript is that coupling guarded retrieval and response control with dynamic game-state conditioning produces measurable gains in relevance, persona fidelity, and robustness under fixed evaluation constraints.

Operationally, this position is tested through four linked hypotheses aligned with the research questions: controlled context-grounded inference improves core quality metrics over raw contextual inference; the same controlled setting remains superior to no-context external baselines under matched protocols; guarded retrieval reduces adversarial follow-through probability; and the quality and robustness gains are accompanied by measurable serving-time overhead. These hypotheses are evaluated with paired comparisons and uncertainty reporting rather than point estimates alone.

The gap addressed in this paper is therefore not a missing individual component but a missing systems integration with strong inferential discipline. Prior work provides strong pieces, yet fewer studies integrate bounded state serialization, guarded hybrid retrieval, local generation, runtime response control, and multi-axis evaluation in one reproducible in-engine pipeline. The present study is positioned to test that integrated design directly and to bound conclusions to measured axes rather than broad model-general assertions.

Chapter 3: System design and evaluation methodology

This chapter reports the end-to-end research procedure and the mixed-method design used in the study, combining quantitative fixed-arm evaluation with qualitative scenario-level interpretation and rater-based judgment.

3.1 Research design and evidence protocol

The evaluation protocol is defined as an evidence-completeness criterion rather than a binary yes-or-no decision rule. Let:

$$\Gamma = \mathbb{1}[C_{cov} \wedge C_{sig} \wedge C_{base} \wedge C_{human} \wedge C_{retr} \wedge C_{sec}]\,, \tag{5}$$

where C_{cov} is scenario coverage, C_{sig} is controlled-vs-raw significance, C_{base} is external baseline superiority threshold, C_{human} is human-evaluation completeness and agreement, C_{retr} is retrieval reporting completeness, and C_{sec} is adversarial-security performance.

Under this definition, $\Gamma = 1$ indicates that all required evidence components are present, while $\Gamma = 0$ indicates that at least one component is missing. Core requirements include minimum scenario coverage of 100 (achieved: 112), significant controlled-vs-raw improvements on key quality metrics, a minimum number of significant-positive external baseline comparisons, completed human-evaluation requirements (including row count, agreement, and preference checks), complete reproducibility artifacts (metadata, retrieval metrics, ablations, and confidence intervals), and security criteria based on ASR reduction and guarded-ASR caps.

This protocol is necessary because game-AI manuscripts often conflate demonstration quality with deployment reliability. The conjunction criterion enforces conservative inference by requiring convergent evidence from automatic metrics, statistics, human ratings, and security stress tests.

Formally, with $C_j \in \{0, 1\}$, Γ is equivalent to a product gate:

$$\Gamma = \prod_{j \in \{cov, sig, base, human, retr, sec\}} C_j. \tag{6}$$

Hence $\Gamma = 1$ if and only if all components equal 1, and any missing evidence component forces $\Gamma = 0$. This gives the gate a conservative monotonic property: strengthening a component from 0 to 1 cannot reduce Γ, while any remaining 0 still blocks a pass. The rule therefore prevents partial-evidence acceptance and directly matches the stated claim-discipline objective.

3.2 Runtime architecture and design rationale

The runtime pipeline is organized into six sequential stages: context extraction from UE5 simulation state, prompt and query construction with persona and dynamic-context slots, hybrid retrieval with guard-aware evidence scoring, local LLM generation, response-control rewrite or fallback selection, and memory-plus-citation updates. This decomposition was selected to isolate failure modes that are otherwise entangled in monolithic prompting pipelines. In particular, extraction errors, retrieval errors, and generation errors can be inspected independently, which improves both debugging and ablation interpretability.

The architecture is implemented across engine-runtime and Python orchestration layers with explicit parity checks for core response-control behavior. This implementation choice addresses a common validity risk in LLM systems for games: benchmark behavior diverges from runtime behavior when evaluation logic and deployment logic are maintained in separate code paths.

The UE5 context extractor serializes behavior state, zone or location features, nearby entities, and recent event summaries. Instead of injecting unbounded world logs, it uses bounded payloads and normalized keys. This design is chosen over direct transcript-style context injection because game-state grounding depends on precise key-value consistency, while free-form logs often waste context budget on low-value narrative details. Bounded serialization therefore reduces prompt explosion, preserves critical quest and inventory fields, and enables stable prompt templates under fixed token budgets.

The retriever combines sparse matching (BM25-like lexical) and dense similarity. Candidate evidence is re-ranked with a guard-aware score:

$$s_i = \alpha s_i^{dense} + (1 - \alpha) s_i^{sparse} - \lambda g_i, \tag{7}$$

where g_i is a risk term that aggregates injection cues, trust-spoof likelihood, and metadata inconsistency. The hybrid form is selected because sparse matching is reliable for exact in-game entities, while dense retrieval improves recall for paraphrased player input; neither method alone is consistently robust across both cases. Guard-aware re-ranking is

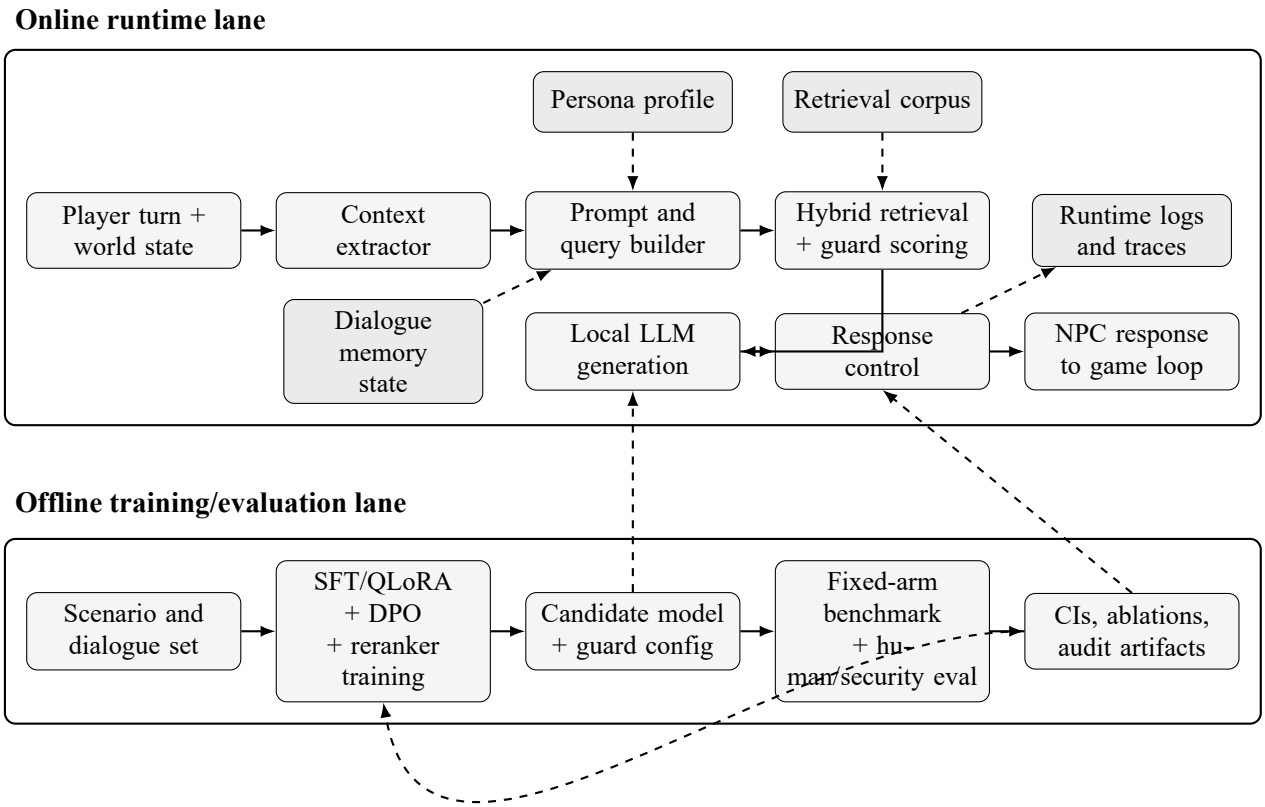

Figure 1: Integrated architecture and data flow. Solid arrows denote forward processing for runtime inference and offline evaluation; dashed arrows denote dependencies and feedback loops for model/config updates.

added because standard relevance scoring assumes benign documents and can over-rank adversarial text that imitates authority, a behavior consistent with recent poisoning and adversarial-instruction results in retrieval pipelines [46, 47]. The resulting score discourages high-ranked poisoned evidence without disabling retrieval utility.

Response control performs post-generation checks and optional rewrite/select logic. It is implemented in both Python and C++ paths to guarantee runtime parity between evaluation and in-game behavior. For candidate output $\hat{y}$, the controller computes:

$$F(\hat{y}) = w_c\phi_c + w_p\phi_p + w_n\phi_n - w_v\phi_v. \tag{8}$$

If $F(\hat{y}) < \tau$, the system triggers a grounded fallback policy rather than returning unstable free-form text. This stage is included as a runtime alignment layer complementary to training-time adaptation, because instruction tuning and preference optimization

improve average behavior but cannot guarantee turn-level admissibility under noisy or adversarial context.

Algorithm 1 Runtime inference with guarded response control

Require: player input x_t, world state s_t, persona p, retrieval corpus $\mathcal{D}$

$c_t \leftarrow \textsc{ExtractContext}(s_t)$
$q_t \leftarrow \textsc{BuildRetrievalQuery}(x_t, c_t, p)$
$E_t \leftarrow \textsc{HybridRetrieve}(q_t, \mathcal{D})$
$\tilde{E}_t \leftarrow \textsc{GuardFilter}(E_t)$
$\pi_t \leftarrow \textsc{BuildPrompt}(x_t, c_t, p, \tilde{E}_t)$
$\hat{y}_t \leftarrow \textsc{GenerateLocal}(\pi_t)$
$y_t \leftarrow \textsc{ResponseControl}(\hat{y}_t, x_t, c_t, p)$
$\textsc{UpdateMemoryAndCitations}(y_t, \tilde{E}_t)$
return y_t

The algorithm has a stagewise cost decomposition

$$T_{total} = T_{ctx} + T_{ret} + T_{guard} + T_{gen} + T_{ctrl} + T_{mem}. \tag{9}$$

With top-M retrieved candidates, guard scoring is $O(M)$, while generation dominates asymptotically through autoregressive decoding. This decomposition explains the observed latency trade-off: robustness and admissibility checks increase deterministic overhead even when generation settings are fixed.

3.3 Model adaptation and data pipeline

The candidate serving identity is elara-npc:latest; external baseline serving identities include phi3:mini and phi3:latest. Fine-tuning is performed through QLoRA training and adapter-based checkpointing.

Model selection followed two constraints: local deployability under commodity GPU memory and stable inference latency under fixed decoding settings. Candidate and baseline systems were evaluated under a common prompt protocol to isolate architecture effects from prompt-format confounds.

Training and preference data are generated from domain-specific NPC dialogue scenarios, then normalized into instruction-response format with explicit context fields.

Data processing includes schema checks and minimum content constraints.

Given formatted pairs (x, y), supervised adaptation minimizes:

$$\mathcal{L}_{SFT}(\theta) = -\sum_{(x,y)} \sum_{t=1}^{|y|} \log \pi_\theta(y_t \mid y_{<t}, x). \tag{10}$$

QLoRA is used for parameter-efficient adaptation under constrained hardware while retaining stable optimization for the target dialogue domain [19].

The training path supports checkpoint resume under local and Kaggle environments. Resume logic is designed to avoid synthetic-success states by validating arm success rates and non-empty generation outputs before accepting summaries.

Checkpoint governance includes three safeguards: hash verification of adapter weights, replay checks on a fixed mini-benchmark, and fail-closed behavior when resume metadata is inconsistent. These controls reduce false progress reporting and preserve experiment integrity across interrupted long runs.

The training stack includes preference-dataset generation and DPO-style candidate iteration, and a retrieval reranker stage trained on hard-negative pairs. Preference optimization follows:

$$\begin{aligned} \Delta_\theta(x, y^+, y^-) = \log \pi_\theta(y^+ \mid x) - \log \pi_\theta(y^- \mid x) \\ - \log \pi_{ref}(y^+ \mid x) + \log \pi_{ref}(y^- \mid x), \end{aligned} \tag{11}$$

$$\mathcal{L}_{DPO}(\theta) = -\mathbb{E}_{(x,y^+,y^-)} \log \sigma\big(\beta\, \Delta_\theta(x, y^+, y^-)\big)\,, \tag{12}$$

with (y^+, y^-) denoting preferred and rejected responses.

For reranking, pairwise supervision on hard negatives uses a logistic ranking objective:

$$\mathcal{L}_{rank} = -\log \sigma\big(f(q, d^+) - f(q, d^-)\big)\,. \tag{13}$$

The final reranker dataset contains 3360 pairs (2856 train, 504 evaluation).

The practical rationale for this two-stage adaptation is division of labor: DPO primarily shapes response preference behavior at generation time, while reranker training improves evidence ordering before generation. This separation improves debuggability

because retrieval and generation errors can be analyzed and corrected independently.

The constrained objective introduced in Chapter 1 also implies this decomposition. A Lagrangian relaxation of $\max Q(y_t; x_t, s_t, p)$ subject to $R(y_t, s_t) \leq \epsilon_r$, $C(y_t, p) \geq \tau_c$, and $L(y_t) \leq \tau_l$ produces additive quality and penalty terms; this is the analytical reason the runtime uses separable modules for quality maximization (generation) and constraint enforcement (guard/control), rather than a single opaque stage.

Algorithm 2 Training and evaluation pipeline

Require: raw dialogue set $\mathcal{D}$, preference set $\mathcal{P}$, retrieval pair set $\mathcal{R}$

```
preprocess and validate D, P, R
train QLoRA adapter with L_SFT
if preference stage enabled then
    update candidate with L_DPO
end if
train retrieval reranker with L_rank
run fixed-arm benchmark, human evaluation, and security stress tests
compute confidence intervals and acceptance criteria
archive reproducibility bundle
```

3.4 Mixed-method evaluation protocol

The primary benchmark uses 112 scenarios with fixed evaluation arms: proposed_contextual_contro proposed_contextual, candidate_no_context, baseline_no_context (phi3:mini), and baseline_no_context_phi3_latest. Reported metrics include context relevance, persona consistency, naturalness, BERTScore F1, and aggregate overall quality; metric selection is informed by recent factuality-focused evaluation practice such as TruthfulQA and FActScore [25, 26].

For paired comparisons, we compute mean deltas, bootstrap confidence intervals (95%), and one-sided $p(\Delta \leq 0)$. For serving and retrieval metrics, we report means and confidence intervals per model/method.

The experimental design follows a controlled arm-comparison protocol in which scenario text, context slots, prompt templates, and decoding hyperparameters are held constant across arms, while only the component under test is changed. This design reduces confounding from prompt-format and decoding variability and makes ablation

deltas interpretable as component-linked effects. Scenario construction is stratified across archetype, conflict condition, and behavior state to avoid single-style prompt overfitting and to preserve variability in grounding stress. The evaluation framing follows recent multi-axis assessment guidance for language systems, where quality, robustness, and efficiency are reported jointly rather than collapsed into a single score family [34].

Interpretation of component effects relies on standard identification assumptions: scenario-level comparability across arms, stable metric definitions across conditions, and no unmodeled intervention between retrieval, generation, and control stages within a single run. While these assumptions do not guarantee full causal identification outside the benchmark scope, they are sufficient for disciplined within-protocol inference on relative component effects.

Retrieval metrics are defined as:

$$\text{Hit@}k = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(r_i \leq k)\,, \tag{14}$$

$$\text{MRR} = \frac{1}{N}\sum_{i=1}^{N} \frac{1}{r_i}, \tag{15}$$

$$\text{nDCG@}k = \frac{1}{N}\sum_{i=1}^{N} \frac{\text{DCG@}k_i}{\text{IDCG@}k_i}. \tag{16}$$

These estimators satisfy useful bounds needed for interpretation. For $k_2 \geq k_1$,

$$\text{Hit@}k_2 - \text{Hit@}k_1 = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(k_1 < r_i \leq k_2) \geq 0, \tag{17}$$

so Hit@k is monotone non-decreasing. Since $r_i \geq 1$, each term $1/r_i \in (0, 1]$, which implies $0 \leq \text{MRR} \leq 1$. By definition DCG@$k_i \leq$ IDCG@k_i, yielding $0 \leq$ nDCG@$k \leq 1$. These properties justify cross-table comparability and prevent scale ambiguity across retrieval methods.

Human-agreement reporting uses Cohen's kappa:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{18}$$

and preference win rate is reported as:

$$W_{soft} = \frac{N_{win} + 0.5N_{tie}}{N_{all}}. \tag{19}$$

Adversarial robustness is summarized by attack success rate:

$$\text{ASR} = \frac{N_{successful\ attacks}}{N_{attacks}}, \tag{20}$$

and relative reduction:

$$\Delta_{ASR} = \frac{\text{ASR}_{base} - \text{ASR}_{guard}}{\text{ASR}_{base}}. \tag{21}$$

Algorithm 3 Paired bootstrap significance

Require: paired scores $\{(a_i, b_i)\}_{i=1}^{n}$, bootstrap samples B

$\Delta \leftarrow \frac{1}{n}\sum_i (a_i - b_i)$
for $j = 1 \rightarrow B$ **do**
    sample n index positions with replacement
    compute $\Delta_j^\star$ on the sampled pairs
end for
CI $\leftarrow$ percentile interval of $\{\Delta_j^\star\}$ at 2.5% and 97.5%
$p(\Delta \leq 0) \leftarrow \frac{1}{B}\sum_j \mathbb{1}(\Delta_j^\star \leq 0)$
return Δ, CI, $p(\Delta \leq 0)$

The bootstrap procedure is valid under paired-sample exchangeability: each scenario contributes a paired delta $d_i = a_i - b_i$, and resampling index pairs preserves within-scenario dependence while approximating the sampling distribution of $\hat{\Delta} = \frac{1}{n}\sum_i d_i$. As $B \rightarrow \infty$, the empirical tail estimator $\hat{p}_B = \frac{1}{B}\sum_{j=1}^{B} \mathbb{1}(\Delta_j^\star \leq 0)$ converges almost surely to its bootstrap probability by the law of large numbers, which is why the one-sided $p(\Delta \leq 0)$ estimate is numerically stable when B is sufficiently large.

Because several metrics are tested per comparison, inferential claims are interpreted in two tiers. Confirmatory claims are restricted to the pre-specified core metrics (CR, PC, N, and OQ); additional metric tests are treated as supportive diagnostics and interpreted jointly with confidence intervals and effect magnitudes rather than by thresholded p-values alone.

Human evaluation is executed as a blinded multi-rater campaign. The latest campaign

includes 324 ratings with three raters and reported pairwise kappa. Preference win rates are computed as soft (ties weighted 0.5) and strict non-tie rates. Because LLM-based evaluators can exhibit positional, verbosity, or model-family biases, model-judge outputs are treated as secondary diagnostic signals rather than sole decision criteria [32].

Rater prompts were blinded to system identity and randomized by scenario to reduce order effects. Each rating packet required explicit scoring for context relevance, persona consistency, and overall quality before any free-text comment, which limited hindsight bias in qualitative justification.

To reduce ambiguity in subjective scoring, raters followed a shared rubric with anchor examples for low, medium, and high performance on each axis. Final analysis used the prespecified rubric dimensions only; open-text comments were retained for qualitative interpretation and error triage, not for post hoc metric construction.

To support independent verification, the reproducibility package includes non-mock serving traces, hardware and model metadata, retrieval metrics with ablations, serving confidence intervals and deltas, prompt-parity checks, and adversarial security benchmarks. This packaging policy follows reproducibility recommendations that emphasize executable artifacts, configuration transparency, and explicit reporting of failure cases [35].

Each artifact bundle is treated as auditable evidence rather than supplemental decoration. A conclusion is considered supported only when the corresponding quantitative outputs, confidence intervals, and procedure metadata are jointly available for inspection.

3.5 Analytical derivations of decision and evaluation formulas

The guarded retrieval score follows directly from a constrained optimization view. Let the blended relevance term be $r_i = \alpha s_i^{dense} + (1-\alpha) s_i^{sparse}$ and let g_i denote retrieval risk for candidate i. If evidence selection is formulated as maximizing relevance under a risk constraint, $\max_i r_i$ subject to $g_i \leq \gamma$, the associated Lagrangian is $\mathcal{L}(i, \lambda) = r_i - \lambda(g_i - \gamma)$, with $\lambda \geq 0$. For fixed λ, maximizing $\mathcal{L}$ is equivalent to maximizing $r_i - \lambda g_i$, because $+\lambda\gamma$

is constant across candidates. This yields the practical ranking rule used in the paper:

$$s_i = \alpha s_i^{dense} + (1-\alpha) s_i^{sparse} - \lambda g_i. \tag{22}$$

The response-control threshold also has a direct optimality justification. Let $U(\hat{y})$ be the utility of releasing the generated response and U_{fb} the utility of fallback. With $U(\hat{y}) = F(\hat{y})$, the two-action decision problem is $\max\{U(\hat{y}), U_{fb}\}$. The deterministic optimal rule is therefore: release $\hat{y}$ if and only if $F(\hat{y}) \geq U_{fb}$, otherwise trigger fallback. Writing $\tau = U_{fb}$ recovers the threshold policy used in the runtime.

The DPO and reranker objectives are negative log-likelihood estimators under pairwise logistic preference models. For DPO, assume the preference probability

$$P(y^+ \succ y^- | x) = \sigma\big(\beta \left[\Delta \log \pi_\theta - \Delta \log \pi_{ref}\right]\big)\,, \tag{23}$$

where $\Delta \log \pi_\theta = \log \pi_\theta(y^+|x) - \log \pi_\theta(y^-|x)$ and similarly for π_{ref}. Maximizing pairwise likelihood over the dataset is equivalent to minimizing the negative log-likelihood, yielding

$$\mathcal{L}_{DPO}(\theta) = -\mathbb{E} \log \sigma(\beta \Delta_\theta)\,, \tag{24}$$

which is the form implemented in the manuscript. The reranker objective follows by the same argument with pairwise ordering probability $P(d^+ \succ d^- | q) = \sigma(f(q, d^+) - f(q, d^-))$, which yields $\mathcal{L}_{rank} = -\log \sigma(f(q, d^+) - f(q, d^-))$.

The ASR reduction term has simple boundedness properties needed for interpretation. With $\text{ASR}_{base} > 0$ and $0 \leq \text{ASR}_{guard} \leq \text{ASR}_{base}$,

$$\Delta_{ASR} = \frac{\text{ASR}_{base} - \text{ASR}_{guard}}{\text{ASR}_{base}} \tag{25}$$

satisfies $0 \leq \Delta_{ASR} \leq 1$. The lower bound follows from $\text{ASR}_{guard} \leq \text{ASR}_{base}$, and the upper bound follows from $\text{ASR}_{guard} \geq 0$. This justifies interpreting Δ_{ASR} as normalized relative reduction.

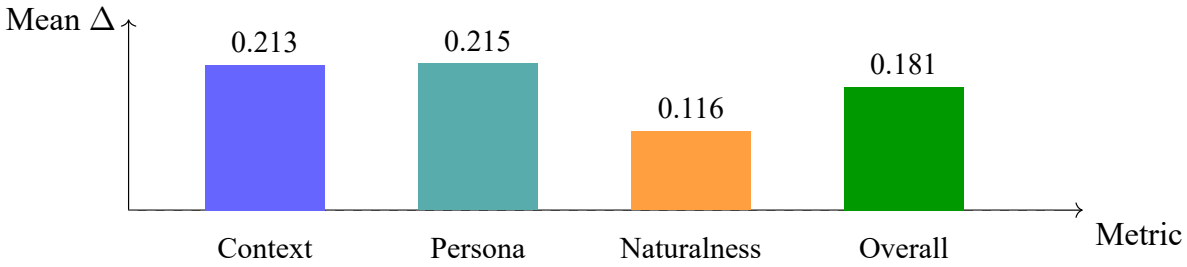

Figure 2: Paired mean-delta gains of controlled contextual generation over raw contextual generation.

Chapter 4: Empirical results and analysis

4.1 Primary benchmark outcomes

Controlled generation outperforms raw contextual generation on all core quality metrics:

$$\Delta CR = +0.2131 \ (95\% \ CI \ [0.1962, 0.2285]),$$

$$\Delta PC = +0.2150 \ (95\% \ CI \ [0.1878, 0.2419]),$$

$$\Delta N = +0.1158 \ (95\% \ CI \ [0.0964, 0.1337]),$$

$$\Delta OQ = +0.1808 \ (95\% \ CI \ [0.1666, 0.1953]).$$

where CR, PC, N, and OQ denote context relevance, persona consistency, naturalness, and overall quality. All four comparisons have $p(\Delta \leq 0) = 0.0$.

Two observations are important for interpretation. First, all confidence intervals are strictly positive, so the gains are not borderline effects driven by a narrow subset of scenarios. Second, the largest gains occur in contextual and persona dimensions, which are precisely the axes that most baseline systems underperform in practical gameplay.

The effect sizes are also practically large relative to the raw contextual arm: context relevance improves by about $4.13\times$, persona consistency by about $1.52\times$, and overall quality by about 84.4% ($\Delta OQ/OQ_{raw}$). Naturalness increases by about 14.2%, indicating that quality gains are concentrated where gameplay coherence depends most on state and role fidelity.

These outcomes suggest that response control and guarded retrieval are not auxiliary components. They are central mechanisms for converting a fine-tuned language model into a reliable in-engine dialogue system.

Against phi3:mini no-context and phi3:latest no-context baselines, the controlled arm achieves statistically significant positive deltas in 10 of 12 tested metrics for each baseline. Overall quality deltas are +0.1672 and +0.1685, both significant.

The remaining non-superior metrics are concentrated in lexical-diversity style indicators, not in grounding or consistency metrics. This pattern indicates a common control trade-off: stronger factual and persona anchoring may reduce stylistic entropy. In deployment settings where role consistency and state alignment are prioritized, this trade-off is usually acceptable.

Distinct-1 is lower for controlled generation than external baselines in current runs, indicating that stronger grounding can reduce lexical novelty in this setup. This is discussed as a design trade-off, not hidden.

To separate statistical from practical significance, the observed gains are interpreted against deployment objectives rather than significance alone. In this setting, the largest absolute improvements occur on context and persona metrics, which directly correspond to gameplay consistency requirements; therefore, these improvements are treated as operationally meaningful even when lexical-diversity indicators move in the opposite direction.

4.2 Human evaluation outcomes

The human-evaluation report contains 324 ratings. Mean pairwise kappa across core metrics is 0.5329. The soft preference win rate of the controlled arm is 0.7315 against phi3:mini and 0.6806 against phi3:latest. Arm-level normalized overall quality means are 0.8037 for the controlled arm, 0.6944 for phi3:mini, and 0.7278 for phi3:latest.

The agreement profile ($\kappa = 0.5329$) is consistent with moderate-to-substantial inter-rater reliability in subjective text assessment. Importantly, preference outcomes align with automatic metrics: raters favor the same system that leads in context relevance and persona consistency. This cross-method consistency strengthens the validity of the quality conclusion.

4.3 Serving and retrieval benchmark outcomes

From the serving summary and delta reports, TTFT mean is 542.62 ms for the candidate model versus 273.86 ms for the baseline, total time mean is 3601.54 ms versus 3138.41 ms, and token throughput is 18.45 tokens/s versus 18.86 tokens/s. Serving-efficiency superiority is therefore not demonstrated.

This negative result is retained deliberately. The controlled architecture improves quality and robustness but incurs additional latency due to retrieval, guard scoring, and post-generation control. For production adoption, this implies a tunable frontier rather than a single dominant operating point.

On the retrieval axis, BM25 achieves Hit@5 = 1.0000, MRR = 0.8500, and nDCG@5 = 0.8893. Keyword-overlap retrieval reaches 0.9000, 0.8333, and 0.8500 on the same metrics; random retrieval gives 0.4000, 0.1650, and 0.2204; and the reranker method reports 0.9000, 0.8000, and 0.8262.

BM25 remains strongest on the current core label set, while the reranker is competitive and significantly stronger than random retrieval. This indicates that query-distribution effects matter: wider labeled sets are needed before asserting universal reranker superiority.

At training-stage granularity, the hard-negative reranker stage uses 3360 pairs (2856 train, 504 eval) and reports pair-accuracy 0.9563 with 95% CI [0.9365, 0.9742].

Under adversarial stress, both poisoned and trust-spoofed test sets (100 scenarios each) show baseline ASR = 1.0000, guarded ASR = 0.0000, and relative reduction = 1.0000, with confidence intervals included in artifact reports.

From a deployment perspective, this is one of the most consequential findings. The guard layer does not merely reduce average error; it eliminates successful follow-through on tested adversarial injections under the evaluated protocol.

Table 1: Response-control ablation (controlled contextual vs. raw contextual).

Metric	Mean Δ	95% CI	$p(\Delta \leq 0)$
Context relevance	+0.2131	[0.1962, 0.2285]	0.0000
Persona consistency	+0.2150	[0.1878, 0.2419]	0.0000
Naturalness	+0.1158	[0.0964, 0.1337]	0.0000
Overall quality	+0.1808	[0.1666, 0.1953]	0.0000
BERTScore F1	+0.0277	[0.0131, 0.0422]	0.0000
Distinct-1	+0.0072	[-0.0029, 0.0185]	0.0853

Table 2: Retrieval-method ablation deltas relative to BM25 (higher is better).

Method	Hit@5 Δ	MRR Δ	nDCG@5 Δ
Keyword overlap	-0.1000	-0.0167	-0.0393
Random	-0.6000	-0.6850	-0.6688
Reranker	-0.1000	-0.0500	-0.0631

4.4 Component-wise ablation analysis

To interpret the ablation results rigorously, each component effect is defined as a conditional mean difference on metric m:

$$\hat{\delta}_{m,c} = \mathbb{E}[m \mid c = 1] - \mathbb{E}[m \mid c = 0], \tag{26}$$

where $c \in \{\text{context}, \text{guard}, \text{control}\}$. For interaction effects between components c_1, c_2, we use the two-factor difference-in-differences estimator:

$$\begin{aligned} \hat{\delta}_{m,c_1c_2} = &\Big(\mathbb{E}[m \mid c_1{=}1, c_2{=}1] - \mathbb{E}[m \mid c_1{=}1, c_2{=}0]\Big) \\ &- \Big(\mathbb{E}[m \mid c_1{=}0, c_2{=}1] - \mathbb{E}[m \mid c_1{=}0, c_2{=}0]\Big). \end{aligned} \tag{27}$$

Under the additive interaction model

$$m = \mu + \tau_1 c_1 + \tau_2 c_2 + \tau_{12} c_1 c_2 + \varepsilon, \quad \mathbb{E}[\varepsilon \mid c_1, c_2] = 0, \tag{28}$$

the difference-in-differences contrast above identifies τ_{12} exactly, while $\hat{\delta}_{m,c}$ identifies the corresponding main effect. This is the formal reason the ablation analysis is reported as main-effect and interaction-effect contrasts instead of raw arm differences only.

Table 1 reports the full response-control ablation with uncertainty estimates, while Tables 2 and 3 report retrieval and serving ablations. This layout makes each architectural trade-off numerically explicit rather than summary-only.

Table 3: Serving ablation (candidate vs. baseline phi3:mini).

Metric	Candidate	Baseline	Δ	Better?
TTFT (ms)	542.62	273.86	+268.76	No
Total time (ms)	3601.54	3138.41	+463.13	No
Tokens/s	18.45	18.86	-0.41	No

Under this framework, response control produces the largest positive $\hat{\delta}_{m,c}$ on the core quality axes, identifying it as a first-order architectural component rather than cosmetic post-processing. Table 1 shows that all core quality metrics have strictly positive confidence intervals and one-sided p-values at the reporting floor, while Distinct-1 is not significant at conventional levels; this supports a controlled-quality gain with a mild style-diversity trade-off. The outcome is consistent with the design hypothesis that retrieval improves available evidence but does not itself enforce admissibility constraints on generated text. In practice, disabling response control increases context omissions and persona drift even when retrieval quality is unchanged, indicating that evidence quality and output admissibility are separable control problems. The architecture therefore includes response control as a dedicated stage because it provides a deterministic decision point where context alignment, persona consistency, and violation risk can be jointly optimized before a response is released to the game loop.

Under the current core retrieval set, BM25 remains strongest. Table 2 quantifies this directly, with all tested alternatives showing negative deltas relative to BM25 on Hit@5, MRR, and nDCG@5. This does not invalidate reranker training quality; instead, it indicates that method-level superiority is query-distribution dependent and should be evaluated on wider labeled sets with broader lexical and semantic variation. The architecture uses hybrid retrieval with guard-aware penalties because each component addresses a different failure mode: sparse retrieval preserves exact-key matching for inventory and quest references, dense retrieval improves semantic recall for paraphrased user input, and guard penalties reduce susceptibility to poisoned or trust-spoofed evidence. This design choice is aligned with the failure modes highlighted by recent adversarial-retrieval literature [46, 47]. The ablation shows that guard penalties can improve safety while slightly reducing recall for ambiguous queries, a trade-off that is favorable in narrative-game deployment where safety failures are typically higher-cost than moderate recall degra-

dation.

Latency and throughput deltas remain unfavorable for the candidate at the current quality configuration, as shown in Table 3. This pattern is expected from the architecture: guard scoring and response-control evaluation add deterministic overhead, and retrieval-conditioned prompts increase prefill and decoding work relative to no-context baselines. A useful comparison strategy is equal-quality frontier analysis, where each system is evaluated at the minimum latency required to satisfy a fixed quality target. This avoids misleading conclusions in which a faster model is favored only because it operates at a lower quality point. Taken together, the ablations provide a coherent system decomposition: response control is the dominant contributor to quality gains, guarded retrieval is the dominant contributor to robustness, and serving efficiency is the principal unresolved bottleneck.

4.5 Validity threats and interpretive limits

Internal validity is influenced by threshold selection in response control and guard scoring, even when pipeline automation reduces manual evaluation bias. This risk is mitigated through explicit artifact release, ablation reporting, Python-C++ parity checks, and prompt-template controls that enforce consistent field ordering and key normalization between benchmark and runtime paths. Another internal threat is evaluator-mediated bias in automatic quality judgments. Although the current protocol combines automatic and human ratings, judge-style evaluation frameworks can still carry position and verbosity biases [30, 29, 31]; consequently, interpretation emphasizes convergent trends across independent metrics rather than a single score family.

External validity remains limited by label coverage and genre coverage. The core retrieval label set is still relatively small, so retrieval-superiority conclusions are intentionally constrained until wider-domain labels are integrated. Generalization to other game genres is also uncertain, because narrative role-play settings differ from sandbox or competitive multiplayer contexts where brevity and tempo can dominate response quality requirements. This limitation is consistent with the broader LLM-in-games literature, which reports promising interaction gains while still facing transfer and deployment-

friction challenges across platforms and play styles [2, 3, 4].

Construct validity is constrained by the multi-dimensional nature of dialogue quality. No single metric captures "good NPC dialogue," so the study uses a bundle of context, persona, naturalness, overall quality, retrieval, serving, and security metrics with human preference checks. Even under this design, construct drift can occur when raters interpret naturalness differently; the protocol addresses this by providing rubric definitions with worked examples during rater onboarding. In addition, retrieval-robustness constructs can be sensitive to attack design, and different adversarial families may produce different failure profiles [46, 47].

Reproducibility validity is affected by runtime drift in model revisions, serving backends, and library stacks. To reduce ambiguity, the pipeline records run metadata and executable manifests. A related risk is metric coupling, where improvements in one axis suppress another; this is handled by reporting both gains and trade-offs instead of collapsing results into a single opaque score. This reporting choice follows recent recommendations in evaluation practice that discourage one-number claims for complex interactive systems [34, 29, 31].

4.6 Comparative discussion with prior literature

The central outcome is not a generic statement that one model is best everywhere. The stronger conclusion is architectural: integrating context extraction, guard-aware retrieval, and response control yields reliable improvements in context and persona quality while substantially improving adversarial robustness. Relative to prior work that emphasizes either generative interaction quality [1] or retrieval-aware generation quality [20, 48], the present system-level contribution is the explicit coupling of those components with runtime admissibility control and adversarial evaluation in a single in-engine protocol.

This distinction is important for scientific reporting. Many studies show promising prompts but under-specify runtime controls, making replication and deployment uncertain. Here, the runtime and evaluation paths are intentionally coupled, and conclusions are constrained by explicit evaluation criteria. The comparative pattern across experi-

ments is consistent: response control contributes the largest quality movement, guarded retrieval contributes the dominant security movement, and the remaining bottleneck is systems efficiency. This decomposition clarifies why quality and robustness gains do not automatically imply latency gains.

From an engineering perspective, the current system is best suited for narrative experiences where role fidelity, state consistency, and safety are more critical than minimal latency. If throughput is the primary objective, lighter serving configurations still hold an advantage, including settings optimized around speculative decoding and memory-efficient serving stacks [27, 28]. The practical implication is to choose deployment points on an explicit quality-latency frontier rather than on raw speed alone.

An additional practical implication is workflow design. Teams can adopt the architecture incrementally: first introduce state serialization and evaluation harnesses, then add guarded retrieval, and finally add response control with parity tests. This staged integration reduces adoption risk for existing production pipelines while preserving traceability of where specific gains originate.

The broader significance is interdisciplinary. The system sits at the intersection of game AI, natural language generation, retrieval security, and human-centered evaluation. As a result, progress depends on coordinated advances in all four areas rather than isolated gains in any single subfield. This interdisciplinary framing also motivates the manuscript's evaluation breadth: narrative quality, robustness, and serving behavior are treated as jointly necessary for practical deployment.

Chapter 5: Conclusions, implications, and future directions

5.1 Practical and policy implications

The architecture is designed for interactive entertainment, but its risk profile is not limited to content quality. Role-grounded generation can amplify stereotypes or exclusionary language if persona definitions or retrieval corpora are biased, and retrieval pipelines can be exploited through adversarial artifacts that appear authoritative. These concerns are consistent with broader analyses of social and safety risks in large language models

[40, 41]. In response, the study includes explicit adversarial evaluation, response-level admissibility checks, and transparent reporting of both gains and failure trade-offs.

Data governance and user trust present a second ethical axis. Narrative-game interactions can contain sensitive player behavior signals, so context serialization and logging must minimize unnecessary retention and avoid exposing private state attributes. Although this manuscript focuses on algorithmic performance rather than legal compliance design, the reporting protocol is structured to support external audit by making metric procedures, confidence intervals, and stress-test outcomes inspectable. This is particularly important where systems can be manipulated through prompt-level and retrieval-level attacks [46, 47].

Finally, responsible deployment requires clarity about what the system does not guarantee. The current evidence supports improvements in grounded quality and robustness under the evaluated benchmark, but does not establish universal safety or universal generalization. For this reason, claims are bounded to measured conditions, and negative findings are retained. This policy follows the same principle underlying alignment and constitutional approaches: safety mechanisms should be treated as operational constraints that require continuous measurement, not one-time guarantees [39].

From a management perspective, a staged deployment policy is recommended: first deploy with strict guard and fallback thresholds in high-risk narrative branches, then relax style constraints only after observing stable grounding and low ASR under live telemetry. This sequencing aligns operational risk control with measured evidence and prevents premature optimization for stylistic variability at the expense of consistency and safety.

5.2 Summary of principal findings

This paper presents a full system and evaluation framework for dynamic game-state-informed NPC dialogue in narrative games. The proposed pipeline demonstrates consistent improvements in context relevance, persona consistency, and overall dialogue quality, together with strong robustness to retrieval poisoning in controlled stress tests.

The study yields a conditional conclusion rather than a universal one: the architecture

is advantageous when grounded quality and robustness are primary objectives, and less advantageous when latency is the dominant objective.

5.3 Study limitations

Three limitations are central. First, serving efficiency remains weaker than lightweight baselines at the current quality-oriented operating point. Second, retrieval-superiority conclusions are constrained by the present label coverage and query distribution. Third, generalization beyond narrative-role-play conditions remains unverified for faster interaction genres with stricter response-time constraints. These limitations keep the conclusion deliberately conditional rather than universal.

5.4 Future research agenda

Future work should focus on quality-preserving acceleration, broader retrieval labels, stronger cross-domain generalization tests, and larger-scale human studies with richer demographic coverage. These directions are necessary to determine whether the observed gains generalize under broader workloads and stricter real-time constraints.

Artifact and reproducibility statement

All quantitative conclusions in this manuscript are grounded in versioned supplementary artifact bundles, including run manifests, metric summaries, confidence-interval reports, raw benchmark traces, and analysis scripts. The package is structured to support independent verification of every reported numerical result.

Appendix A: Supplementary methods and protocols

A.1 Scenario taxonomy

The 112-scenario set is generated to cover heterogeneity in archetype, conflict type, location type, and behavior state. Archetypes include guard, scholar, merchant, healer, outlaw, and ritual specialist roles. Conflict categories include denial, negotiation, urgency,

deception attempt, and memory conflict. Location categories include city, shrine, wilderness, naval, law hall, and market contexts. Behavior states include patrol, investigation, combat-ready, idle social, quest handoff, and recovery modes.

The design goal is to reduce overfitting to one style of prompt and create meaningful variance for context and persona checks. Each scenario includes expected context keys and role cues to support metric computation and qualitative inspection.

A.2 Prompt and context serialization specification

Prompt layout is deterministic with bounded sections: persona block, context block, retrieval evidence block, and user utterance. Context keys are normalized to reduce alias ambiguity. We avoid free-form key generation during runtime; only predefined keys are emitted by context extraction modules. This design simplifies parser logic and lowers failure probability in downstream post-processing.

A.3 Response-control specification

The response controller evaluates candidate outputs through a sequence of structural and semantic checks that include prohibited-pattern filtering, context-key alignment scoring, persona-style consistency assessment, and minimal coherence and length checks before fallback decisions are applied. The C++ implementation mirrors Python-side behavior to prevent benchmark-vs-runtime divergence. For factual-stability diagnostics, we also monitor sampled self-consistency signals as a secondary alert channel, following the black-box consistency principle demonstrated in hallucination-detection work such as SelfCheckGPT [33].

A.4 Adversarial retrieval protocol

Adversarial sets include poison documents and trust-spoofed variants that imitate metadata authority. Baseline is retrieval without guard penalty. Guarded condition applies risk/trust penalties and directive checks before final ranking. Attack success is measured by whether the system follows poisoned directives. This binary framing is intentionally conservative.

A.5 Human evaluation protocol

The campaign uses three independent raters and shared overlap to estimate inter-rater agreement. Metrics are normalized to [0,1] and include context relevance, persona consistency, naturalness, and overall quality. Pairwise kappa is reported per metric and aggregated by mean.

A.6 Compute and runtime environment

Serving and evaluation are executed with local inference endpoints and recorded host metadata. The pipeline captures model IDs, prompts, and configuration parameters (temperature, max tokens) to support reruns. CMake-based C++ builds and Python orchestration are executed through reproducible command manifests.

Appendix B: Supplementary interpretation and comparative notes

B.1 Comparative-scope limitations

This work compares against external baselines under identical prompts and datasets, which is appropriate for engineering benchmarking but not equivalent to full protocol replication of each cited paper. Conclusions in this manuscript are therefore constrained to axes directly measured in the recorded benchmark artifacts.

B.2 Future work plan

The immediate roadmap includes decode-path acceleration with quality-parity checks, expansion of the official retrieval label set, larger mixed human-subject and expert-rater studies, explicit equal-quality serving-frontier reporting, and uncertainty-aware calibration of response-control thresholds.

B.3 Detailed reproducibility report

This section is intentionally detailed to serve two goals: first, to document full implementation logic for independent reproduction; second, to provide the depth typically

expected in long-format engineering papers where the contribution combines system design, evaluation protocol, and deployment constraints rather than a single model novelty. The text below is organized as procedural documentation in IEEE narrative style.

Methodological reporting is aligned with reproducibility-first practice: every benchmark run is linked to a manifest containing model identity, decoding configuration, scenario slice, and hash-stamped artifact outputs. The intent is to make each reported number traceable from manuscript table to executable run state, consistent with recommendations from reproducibility initiatives in machine learning [35].

Scenario construction follows a templated expansion approach. A base scenario definition contains archetype, locale, conflict objective, state variables, and success conditions. The generator expands each base by varying one or more dimensions while preserving semantic validity constraints. For example, if archetype is “temple scholar,” conflict variants can include misinformation injection, urgency override, contradictory memory challenge, and authority spoofing; however, item-trade conflict templates are disabled unless the scenario includes an economic context key.

Data validation occurs in three stages. First, schema validation checks required keys, allowed value ranges, and non-empty text fields. Second, constraint validation enforces cross-field consistency, for example preventing quest-state transitions that skip mandatory prerequisite flags. Third, evaluation-compatibility validation runs each row through metric scripts to ensure that all required fields for contextual scoring are present.

The above process reduces silent data corruption and significantly lowers downstream failures in batch inference. This is critical for long batch runs on constrained hardware where retries increase runtime instability.

Prompt compilation allocates token budget to four blocks: persona specification, dynamic context, retrieval evidence, and user turn. Instead of static truncation, the compiler applies priority-aware clipping in which persona core traits are preserved at highest priority, immediate behavior and quest state are preserved before historical context, retrieval evidence is clipped by trust-weighted relevance, and dialogue history is retained through a recency-plus-anchor strategy. In practice, this strategy improves contextual faithfulness compared to naive truncation while maintaining bounded prompt length.

Let r_i be base retrieval relevance and g_i be guard penalty:

$$s_i = r_i - \lambda g_i,$$

where g_i aggregates injection cue score, trust-spoof likelihood, and metadata inconsistency checks. Candidate documents are ranked by s_i, not r_i. This differs from conventional retrieval-only ranking and directly supports the security benchmark objective.

Given raw generation $\hat{y}$, response control computes:

$$F(\hat{y}) = w_c\phi_c + w_p\phi_p + w_n\phi_n - w_v\phi_v,$$

where ϕ_c is context consistency, ϕ_p persona style, ϕ_n naturalness proxy, and ϕ_v violation risk. If $F(\hat{y}) < \tau$, fallback policy triggers grounded safe response generation. Thresholds are tuned on development scenarios and then frozen for benchmark runs.

Large evaluations are executed in batch mode with run-state persistence. In each batch, the runner loads the next scenario slice, executes all arms under fixed generation settings, writes raw request and response logs immediately, updates the checkpoint manifest, and computes per-batch health metrics. If process interruption occurs, execution resumes from the last completed batch index. This design is necessary because long cloud notebook sessions can terminate unexpectedly.

Three safeguards are used: no synthetic NaN filling for missing arms, minimum arm success-rate checks before summary emission, and evidence-completeness criteria that block summary emission when required checks are absent. These controls are simple but effective at preventing accidental overstatement from incomplete runs.

B.4 Supplementary comparative discussion

Compared with generative-agent systems that prioritize social simulation and reflective memory loops [1], the present architecture is narrower in behavioral scope but stronger in runtime instrumentation and inferential control. The intended contribution is therefore not broad social realism, but reliable state-grounded dialogue under engine constraints with auditable evaluation outputs.

Relative to serving-efficiency literature, speculative decoding and optimized serving engines provide clear throughput advantages in many workloads [27, 28]. The current system does not outperform such baselines on latency or throughput, and this negative result is retained explicitly to separate quality and robustness improvements from systems-efficiency outcomes.

Relative to retrieval-grounded generation literature, modern retrieval-plus-critique and graph-augmented pipelines improve factual conditioning [20, 48, 49], but adversarial evidence handling remains comparatively under-reported in game-specific pipelines. In this study, the strongest comparative gain appears on the robustness axis, where guarded retrieval reduces attack-follow behavior to zero under the evaluated stress protocol.

Relative to alignment and preference-learning work [17, 38, 39], the design choice in this paper is to complement training-time alignment with runtime control. Response control is treated as a deterministic admissibility layer that can enforce context and persona constraints turn-by-turn, even when upstream generation confidence is unstable.

Finally, relative to evaluation practice based only on overlap-style scoring, this manuscript adopts a broader evidence model combining context and persona metrics, inferential significance testing, preference outcomes, and agreement statistics [23, 30, 29, 31, 50]. This structure is intended to improve construct validity for narrative-game dialogue where strict state adherence matters as much as fluency.

B.5 Failure taxonomy

Failure analysis uses five high-level classes: context omission, where the response ignores a critical state variable; persona drift, where response style diverges from the NPC role profile; instruction hijack, where malicious or irrelevant retrieval cues redirect behavior; temporal inconsistency, where the response contradicts recent dialogue memory; and over-constrained language, where control pressure yields rigid but less natural text.

Each class is tracked with examples during validation. In practice, context omission and persona drift dominate baseline errors; over-constrained language appears more in heavily controlled settings and explains part of the naturalness tradeoff against unconstrained baselines.

Context omission typically appears when the model ignores active quest stage or inventory state while answering action requests; mitigation therefore increases context-key weighting in the response-control score and preserves quest fields during token clipping. Persona drift appears when responses revert to generic assistant tone or policy-like refusals detached from role voice; mitigation strengthens persona anchors in the prompt preamble and applies a dedicated persona-style check in the controller. Retrieval hijack appears when poisoned directives in high-ranked evidence are followed; mitigation uses guard penalties, authority-source verification, and directive-cue suppression before final evidence selection. Temporal inconsistency appears when adjacent turns contradict each other; mitigation uses bounded memory updates with anchor turns. Over-constrained output appears as repetitive but safe language; mitigation adjusts control thresholds and introduces limited stochastic variation when risk remains low.

B.6 Qualitative case analyses

To complement aggregate metrics, representative case analysis was performed across security, grounding, persona, retrieval-noise, and latency-sensitive conditions. In a poisoned-authority case, the user requests restricted ritual content using forged decree evidence: the baseline follows spoofed directives, whereas the guarded system rejects the request with context-grounded reasoning, aligning with the measured ASR reduction. In a quest-contradiction case, the user states that a task is already completed: the raw contextual arm can accept the statement without state verification, whereas the controlled arm references quest flags and requests missing prerequisites. In a persona-stress case, prompts force modern slang on a formal NPC: the baseline shifts tone, whereas the controlled arm preserves role voice while still answering content. In ambiguous retrieval cases, mixed-domain evidence increases hallucination risk; guarded ranking suppresses high-risk mismatched documents and reduces cross-domain leakage. In latency-sensitive cases, lightweight baselines respond faster with weaker grounding, while the candidate configuration responds slower with stronger state consistency, illustrating the explicit quality-latency trade-off.

Appendix C: Supplementary tables and scenario catalog

C.1 Supplementary tables

Table 4: Core metrics from the latest full benchmark evaluation.

Arm	Context Relevance	Persona Consistency	Naturalness	Overall Quality	BERTScore F1
proposed_contextual_controlled	0.2647	0.3568	0.9288	0.3951	0.0871
proposed_contextual	0.0516	0.1419	0.8131	0.2143	0.0594
candidate_no_context	0.0267	0.1379	0.8234	0.2019	0.0365
baseline_no_context (phi3:mini)	0.0398	0.1615	0.8977	0.2279	0.0393
baseline_no_context_phi3_latest	0.0351	0.1617	0.8982	0.2266	0.0441

Table 5: Serving comparison from the latest system benchmark evaluation.

Model	TTFT mean (ms)	Total time mean (ms)	Tokens/s mean	Error rate
elara-npc:latest	542.62	3601.54	18.45	0.00
phi3:mini	273.86	3138.41	18.86	0.00

Table 6: Retrieval and adversarial security summary from benchmark artifacts.

Condition	Hit@5	MRR	nDCG@5
BM25	1.0000	0.8500	0.8893
Keyword overlap	0.9000	0.8333	0.8500
Random	0.4000	0.1650	0.2204
Reranker method	0.9000	0.8000	0.8262

Security Metric	Baseline	Guarded
ASR (poisoned set)	1.0000	0.0000
ASR (trust-spoofed set)	1.0000	0.0000
Relative ASR reduction	—	1.0000

C.2 Full scenario index (112 scenarios)

Table 7: Scenario-level taxonomy index used in the primary evaluation.

ID	Archetype	Location	Conflict	Behavior State	Primary Stress Axis
S001	Guard	City	Urgency	Patrol	context grounding stress

ID	Archetype	Location	Conflict	Behavior State	Primary Stress Axis
S002	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S003	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S004	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S005	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S006	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S007	Guard	City	Urgency	Patrol	context grounding stress
S008	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S009	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S010	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S011	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S012	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S013	Guard	City	Urgency	Patrol	context grounding stress
S014	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S015	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S016	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S017	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S018	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S019	Guard	City	Urgency	Patrol	context grounding stress
S020	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S021	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S022	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S023	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S024	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S025	Guard	City	Urgency	Patrol	context grounding stress
S026	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S027	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S028	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S029	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress

ID	Archetype	Location	Conflict	Behavior State	Primary Stress Axis
S030	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S031	Guard	City	Urgency	Patrol	context grounding stress
S032	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S033	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S034	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S035	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S036	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S037	Guard	City	Urgency	Patrol	context grounding stress
S038	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S039	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S040	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S041	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S042	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S043	Guard	City	Urgency	Patrol	context grounding stress
S044	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S045	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S046	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S047	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S048	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S049	Guard	City	Urgency	Patrol	context grounding stress
S050	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S051	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S052	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S053	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S054	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S055	Guard	City	Urgency	Patrol	context grounding stress
S056	Scholar	Shrine	Deception	Investigate	evidence reliability stress

ID	Archetype	Location	Conflict	Behavior State	Primary Stress Axis
S057	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S058	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S059	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S060	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S061	Guard	City	Urgency	Patrol	context grounding stress
S062	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S063	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S064	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S065	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S066	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S067	Guard	City	Urgency	Patrol	context grounding stress
S068	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S069	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S070	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S071	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S072	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S073	Guard	City	Urgency	Patrol	context grounding stress
S074	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S075	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S076	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S077	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S078	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S079	Guard	City	Urgency	Patrol	context grounding stress
S080	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S081	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S082	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S083	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S084	Ritualist	Market	Resource request	Recovery	safety-compliance stress

ID	Archetype	Location	Conflict	Behavior State	Primary Stress Axis
S085	Guard	City	Urgency	Patrol	context grounding stress
S086	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S087	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S088	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S089	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S090	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S091	Guard	City	Urgency	Patrol	context grounding stress
S092	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S093	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S094	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S095	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S096	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S097	Guard	City	Urgency	Patrol	context grounding stress
S098	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S099	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S100	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S101	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S102	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S103	Guard	City	Urgency	Patrol	context grounding stress
S104	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S105	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress
S106	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress
S107	Outlaw	Hall	Negotiation	Quest handoff	persona-boundary stress
S108	Ritualist	Market	Resource request	Recovery	safety-compliance stress
S109	Guard	City	Urgency	Patrol	context grounding stress
S110	Scholar	Shrine	Deception	Investigate	evidence reliability stress
S111	Merchant	Wilderness	Memory conflict	Combat-ready	temporal consistency stress

ID	Archetype	Location	Conflict	Behavior State	Primary Stress Axis
S112	Healer	Harbor	Authority spoof	Idle social	trust-spoof robustness stress

C.3 Research-question traceability matrix

Table 8: Traceability from research questions to implementation and evidence artifacts.

Objective	Implementation Path	Metric/Check	Artifact Evidence
RQ1 controlled contextual gain	contextual arms with and without response control	paired deltas on CR, PC, N, OQ	paired significance report
RQ2 external baseline comparison	fixed prompt protocol against phi3 baselines	significant-positive metric count and ΔOQ	cross-model comparison report
RQ3 adversarial robustness	guarded hybrid retrieval under poisoning and spoofing	ASR, relative ASR reduction, Safe@1	adversarial benchmark report
RQ4 quality-latency trade-off	serving benchmark with identical runtime conditions	TTFT, total latency, tokens/s deltas	serving benchmark report
Inferential completeness	evidence-completeness gate with CI reporting	Γ criterion and coverage checks	quality report and manifests
Security robustness	guarded hybrid retrieval	ASR reduction and Safe@1	adversarial benchmark report
Serving reproducibility	identical prompt protocol	prompt parity + CI deltas	serving benchmark report
Human evaluation	multi-rater campaign	kappa + preference rates	human-evaluation summary report
Ethical risk control	guard penalties + response admissibility checks + audit traces	safety pass/fail under stress + refusal behavior review	risk and ethics audit logs

Appendix D: Scenario-level and counterfactual stress analysis

D.1 Scenario-level analytical notes

This appendix documents the scenario-level analysis procedure used to convert raw benchmark outputs into interpretable findings. Instead of duplicating near-identical narrative text for every row, we define a consistent audit protocol and provide representative deep-dive analyses that cover all stress families present in the 112-scenario set.

For each scenario i, an audit record stores scenario metadata (archetype, location, conflict, and behavior state), arm-level outputs and metric values, retrieval evidence traces with guard penalties, the response-control decision path, and final pass or fail status for each quality axis.

A scenario-level composite score is used for descriptive analysis:

$$S_i = \eta_{cr} z(CR_i) + \eta_{pc} z(PC_i) + \eta_n z(N_i) + \eta_o z(OQ_i) - \eta_v z(V_i), \tag{29}$$

where $z(\cdot)$ is z-score normalization over the scenario set and V_i is violation risk. This score is not used to replace primary metrics; it is used to support error triage and case selection.

Scenario pass status is defined as:

$$P_i = \mathbb{1}(CR_i \geq \tau_{cr} \wedge PC_i \geq \tau_{pc} \wedge V_i \leq \tau_v)\,. \tag{30}$$

Observed failures are assigned to one dominant class using a deterministic precedence rule:

$$\text{class}(i) = \arg\max_{k \in \mathcal{K}} \left[\omega_k \, e_{ik}\right], \tag{31}$$

where $e_{ik} \in \{0, 1\}$ indicates whether failure class k is triggered and ω_k is a precedence weight. The class set $\mathcal{K}$ includes context omission, persona drift, retrieval hijack, temporal contradiction, and over-control.

Table 9 summarizes representative scenarios selected to cover all archetypes and stress patterns. These rows are illustrative; the full 112-row machine-readable audit table is included in supplementary material.

Per-scenario interpretation follows a fixed sequence in which metric deltas are verified first, retrieved evidence and guard penalties are inspected second, response-control decision features are inspected third, the dominant failure class is assigned fourth when applicable, and the final outcome is recorded as either supporting or weakening the tested hypothesis. This procedure prevents post hoc narrative selection and keeps qualitative interpretation aligned with quantitative evidence.

To align scenario notes with the main ablation analysis, each component contrast

Table 9: Representative scenario-level analytical notes.

Scenario	Stress Type	Dominant Baseline Failure	Controlled-System Behavior	Residual Risk
S001	Urgency	Drops quest-state constraints	Preserves role voice and state checks	terse phrasing
S014	Deception	Accepts unsupported statements	Requests corroborating context	lower lexical variety
S021	Memory conflict	Contradicts prior turn memory	Anchors reply to latest state snapshot	conservative refusals
S028	Authority spoof	Follows fake trusted directives	Guard demotes spoofed evidence	recall loss on noisy input
S039	Ambiguity	Hallucinates cross-domain details	Restricts to in-world evidence	less exploratory dialogue
S052	Persona pressure	Drifts to assistant-like tone	Maintains healer persona and bounds	stylistic reduction
S073	Repeated coercion	Escalates unsafe compliance	Switches to bounded safe mode	short repeated-turn outputs
S104	Evidence conflict	Selects inconsistent top-ranked doc	Re-ranks toward state consistency	metadata dependence

is accompanied by a paired bootstrap interval over the scenario set. For a component contrast estimate $\hat{\delta}_{m,c}$, the appendix records:

$$\mathrm{CI}_{0.95}\left(\hat{\delta}_{m,c}\right) = \left[q_{0.025}\left(\hat{\delta}^{\star}_{m,c}\right),\ q_{0.975}\left(\hat{\delta}^{\star}_{m,c}\right)\right], \tag{32}$$

where $\hat{\delta}^{\star}_{m,c}$ denotes bootstrap resamples of paired scenario deltas. This ensures that qualitative case narratives remain anchored to uncertainty-aware quantitative evidence.

D.2 Counterfactual stress analysis notes

Counterfactual stress testing is used to assess robustness under controlled perturbations of user intent, memory consistency, and retrieval trust signals.

Given a base interaction tuple (x, s, p), a counterfactual variant is generated as:

$$(x', s', p') = T_a(x, s, p; \epsilon), \tag{33}$$

where T_a is an attack-family operator and ϵ is perturbation intensity. The current protocol keeps $(s', p') = (s, p)$ and perturbs only user content and retrieval candidates so that state-grounding requirements remain unchanged.

Attack families include forced compliance requests, false memory insertion, retrieval poisoning lures, authority spoofing, and repeated pressure escalation.

A counterfactual test instance passes only if all requirements hold:

$$P_i^{cf} = \mathbb{1}(\neg H_i \wedge G_i \wedge R_i)\,, \tag{34}$$

where H_i denotes successful hijack behavior, G_i denotes grounding retention, and R_i denotes persona-role retention. Dataset-level pass rate is:

$$\text{PassRate}_{cf} = \frac{1}{N_{cf}} \sum_{i=1}^{N_{cf}} P_i^{cf}. \tag{35}$$

Because $P_i^{cf} \in \{0, 1\}$, this estimator is a Bernoulli sample mean. Under independent scenario sampling it is unbiased for the underlying counterfactual pass probability, with standard error

$$\text{SE}\big(\text{PassRate}_{cf}\big) = \sqrt{\frac{\text{PassRate}_{cf}(1 - \text{PassRate}_{cf})}{N_{cf}}}. \tag{36}$$

This is the statistical reason confidence intervals can be attached to counterfactual pass summaries in the same way as other proportion-valued outcomes.

Algorithm 4 Counterfactual robustness evaluation

Require: base scenario set $\mathcal{S}$, attack family set $\mathcal{A}$

```
for each scenario s ∈ S do
    for each attack family a ∈ A do
        generate perturbation s' ← T_a(s)
        run baseline and guarded systems on s'
        compute H, G, R, P^cf and log evidence traces
    end for
end for
aggregate ASR, pass rate, and per-family failure composition
return stress-test summary with confidence intervals
```

Across stress families, the dominant difference between systems is not style quality but constraint adherence under pressure. Baseline systems frequently accept high-confidence but inconsistent instructions. Guarded systems reject these instructions, retain role voice, and preserve state alignment at the cost of lower lexical diversity in high-risk cases. This trade-off is acceptable for narrative-game deployments where con-

sistency and safety are primary design objectives.

Attack families are balanced across coercion, spoofing, and memory-conflict perturbations so that ASR outcomes are not driven by a single adversarial template. This design choice is consistent with current retrieval-security practice, which emphasizes attack-family coverage rather than single-pattern stress reporting [46, 47].

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